Modeling Communities in Peril

Polarization and the changes in the social fabric of congregations through an agent-based approach

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If we are united in Christ, why so much division?

A man was rescued after twenty years alone on a desert island. The sailors found three huts on the beach. "What are these?" they asked. "That one is my house. That one is my church." "And the third?" "Oh, that's the church I used to go to." Funny, until you notice it is our story.

Jesus prayed for those who would believe in him "that they may all be one." If we are united in Christ, why is there so much division?

In 1980, 96 of every 100 Brazilians were Christian. By 2022, it was 84. In the same years, evangelical churches multiplied. One study of the national business registry counted 109,560 of them in 2019, with 6,356 new ones opened that year: about seventeen a day. The share of Christians fell while the number of churches grew. Many of those churches are mission work. Some are splits.

When I joined Calvin University, I brought with me a question that had been following me since my years in Brazil: what makes people stay in a community when the social pressure around them pulls in a different direction? I had witnessed firsthand how political polarization can fracture evangelical communities: how networks once used to share devotionals became vehicles for disinformation, and how the fragmentation of a shared social space could unravel bonds that had seemed permanent. That experience gave me both a question and a vocation: could computational agent-based modeling help us understand, and perhaps mitigate, the forces that fragment the body of Christ?

1. Why now: the sorted church

The United States shows the same strain. Ryan Burge pastored a small Baptist church in Mount Vernon, Illinois, for seventeen years. In the 1960s more than three hundred people worshiped there. By 2006 it was about fifty. In its last year, some Sundays had eight or ten. It closed in 2024.

Ryan Burge's data show that the long rise of Americans with no religion has leveled off. Churches have kept members by becoming politically uniform. In Burge's words, "white Christianity is becoming more and more politically homogeneous with every passing year." The congregation where liberals, conservatives, and moderates sat side by side on Sunday morning is becoming rare.

This division can be seen from the other side of the bridge also. Black Protestants are just as uniform in the other direction: about 95 percent voted for Obama, and in 2024 only about 12 percent voted for Trump. Liberals who could not find a congregation that fit them mostly left. In 2008, 28 percent of Democrats claimed no religion. By 2021 it was 45 percent. Burge calls the nones "the default landing spot for political liberals who can't find a place in American Christianity."

Polarization is also often misdiagnosed. The real challenge emerges when people lose the capacity to share the same space with those who hold different perspectives. It is this breakdown of coexistence, not difference itself, that produces division, friction, and conflict.

I work to try to understand what happens inside a congregation when politics pulls on the same thread as faith. Is a uniform congregation stronger, or only quieter?

2. AI is bigger than generative AI

To explain my field of studies, I want to start by reclaiming a misused term: artificial intelligence.

Say "artificial intelligence" today and most people picture a machine that writes, draws, and talks back, such as ChatGPT or Claude. Large Language Models and generative AI are branches of a much older tree. AI is a big tree where you have branches such as machine learning, planning, reasoning, vision, robotics, and the branch I work on: multi-agent systems.

Lately the industry has fancied the term "agentic AI." That is a little awkward for those of us who have been building agents for decades. The standard textbook in the field defined artificial intelligence as the study of agents in 1995. We are glad for the company. We would only like our word back.

Agent-based modeling (ABM) is a subfield of artificial intelligence that studies emergent phenomena in different aspects of life by building them from the bottom up. A population of software agents, each with its own state and its own rules, is generated and set loose in

a simulated environment, while the researcher observes the results of the collective interactions between agents, and between agents and the environment.

My agents go to church, talk to their neighbors, and some of them stop coming.

I face this work as a craft rather than a method because none of its steps is mechanical. The modeler decides which features of a person survive into an agent and which are left behind, how agents meet and influence one another, what counts as a valid run, and what the pattern on the screen is allowed to mean. Each of those decisions is a reduction, made by hand, at the modeler's expense.

Building a model of the church is a series of decisions about what a person is, and those decisions are epistemological and theological.

3. Two ways of modeling religion

Religion has been simulated for about two decades: ritual, religious violence, secularization, the spread of unbelief.

Much of social computing research proceeds within a naturalistic framework, assuming that social phenomena are fully explicable through observable causes and measurable variables, without reference to anything beyond the natural order. Representative of this tendency is the computational modeling of religion pursued by Wesley Wildman and LeRon Shults, which traces the transformation of worldviews across civilizations through agent-based simulation, a powerful methodology that nevertheless explicitly brackets transcendence as an explanatory category.

This is not a neutral methodological choice. It is a worldview, and it shapes what the models are allowed to find. A Christian Reformed epistemology insists that this exclusion is not a simplification. It is a distortion.

Theology has modeled the church for much longer. In 1974 Avery Dulles described the church through five models: institution, mystical communion, sacrament, herald, and servant; he later added a sixth, the community of disciples. Dulles's models are images for a mystery. They describe what the church is, and say little about how a congregation behaves on an ordinary Sunday, or why some of its members stop coming.

The difference lies in the anthropology: what the agents are assumed to be, which is written into their rules. A framework that brackets transcendence tends to model religious commitment as a cost-benefit calculation or an identity marker, one axis among

the agent's attributes. A Reformed anthropology requires that different aspects, such as belief, behavior, and belonging, be held together as dimensions of a single person, none of which can stand in for the others.

So the model I will describe takes its method from the first tradition and its anthropology from the second.

4. Belief, behavior, belonging: the craft in practice

The specific framework I have developed, the Trifold Community Model, represents congregational adherence through three factors: belief, behavior, and sense of belonging. These dimensions reflect a Reformed understanding of personhood, what theologians call an anthropology, in which faith is never merely a matter of the head, but is always lived out in practices and embedded in community. The choice of these three dimensions is itself a theological statement: persons cannot be reduced to their beliefs alone, nor to their attendance patterns, nor to their social ties in isolation. Within this framework, it is the interplay of all three that stands for a person in community.

Behavior shapes belonging. Belief shapes behavior. And belonging shapes belief. And the other way around also. Polarization is not merely a problem of wrong beliefs requiring correction, nor simply a behavioral dysfunction amenable to communication training. It is a triadic wound: to one's sense of belonging, to one's embodied habits of relationship, and to one's capacity for open-minded inquiry.

Three numbers stand for a person here. That is a reduction we need to acknowledge. What the model won't permit is to let one of the three stand for the whole. A person who attends faithfully but has grown alienated in their sense of belonging is not the same as one whose attendance has lapsed; a person whose political beliefs have drifted is not the same as one whose religious commitment has weakened. The distinction matters, both pastorally and computationally.

The community dynamics that emerge (polarization, resilience, fragmentation) arise from the local interactions of these particular persons, not from rules imposed at the community level. Diversity is not noise to be controlled for, but the structure through which social reality unfolds.

The version we are building now makes three more decisions of this kind. First, belonging is not a dial the model turns. It is read off the web of relationships a person actually has, because relationships are part of what a person is. Second, the model keeps two questions apart: whether listening warms our feelings toward those unlike us, and

whether it changes what we believe. The model has to test the second. Third, it adds one mechanism at a time, and it lists what it leaves out, so that no one mistakes an omission for an oversight.

Bavinck reminds us that, in God, "there is unity in diversity, diversity in unity"; among creatures "we see only a faint analogy of it." "Among us unity exists only by attraction, by the will and the disposition of the will; it is a moral unity that is fragile and unstable." This, in my opinion, describes a congregation. Its members are held together by nothing stronger than attraction and choice: the decision to attend on Sunday, the sense of belonging that draws one back, the confession shared with others, and every one of them can be withdrawn.

Bavinck also names the two ways such a unity fails, which are also represented in the model. Either "the unity swallows up the diversity" and the congregation grows uniform and brittle, or the diversity outruns the unity and the congregation comes apart. What we seek is unity that does not erase difference, difference that does not dissolve unity.

The model is also built to travel. The two belief axes are set for each place: in the United States the religious axis might run from liberal to conservative theology, in Brazil from Pentecostal to mainline, in Kenya across an interfaith community. The mechanism stays the same; the labels and the starting conditions change.

5. What models could do for the church

The model is still being built, so what follows is a hypothesis to test. Once it is checked against real congregations, a model like this could do two things for a church. It could make visible what a congregation lives through but cannot see: belonging thinning at the edges before anyone stops coming. And it could let leaders ask "what if" before living it: what happens to a congregation if tension keeps rising, or if more of its members learn to listen well.

6. What the model cannot do, and why

Lewis Smedes offered a framing that has stayed with me: in our fallen world, we believe we are meant "to go into the world and create some imperfect models of the good world to come." The word imperfect matters. Smedes had in mind lived communities that prefigure, however faintly, the world to come. My models are something humbler: scientific representations of the world as it now is, fallen and fragmented. They are built for the world to come.

The more accountable a model is to the communities it studies, the more clearly it reveals what it cannot represent. Our model illuminates dynamics of polarization and resilience, but it is silent about grace. It cannot represent the elder who stays when others leave, whose quiet faithfulness anchors a congregation that every metric would have predicted would collapse. It cannot measure the effect of a sermon that lands differently on a community that has just passed through a crisis than on one that has not. It cannot account for prayer, nor for the movement of the Holy Spirit.

This silence is deliberate. It is a model that works within its scope. What it represents is the unity creatures make for themselves, the unity that in Bavinck's words exists "only by attraction, by the will" and is "fragile and unstable"; what it cannot represent is the unity the church receives rather than produces.

What we do not measure is not thereby unimportant. It may be, as Bavinck would insist, the most important thing of all.

7. An invitation

Polarization is not going away. The diversity of human thought and experience will always generate friction and social tension. But friction does not have to become fracture.

Models of the church cannot be built well by any one discipline. I am asking theologians, social scientists, pastors and modelers to work at the same table: help shape the assumptions, bring your congregations' questions, and tell us where the model is wrong.

Extensive ethnographic fieldwork will precede the computational modeling, reflecting the order Bavinck insists on: we do not write laws onto the communities we study. We listen first, and build models that are accountable to what we hear.

Those who build these models build imperfect ones. They listen before they build. And I trust that this work, partial, fallible, reformed and always reforming, is one way of taking up the charge Smedes gave: "to go into the world and create some imperfect models of the good world to come." John saw that world once, and what he saw was not uniformity: "a great multitude that no one could number, from every nation, from all tribes and peoples and languages, standing before the throne and before the Lamb" (Revelation 7:9). That is the unity worth wanting, and it keeps every difference it gathers. Every model I build of this world is built for that one, the good world to come.